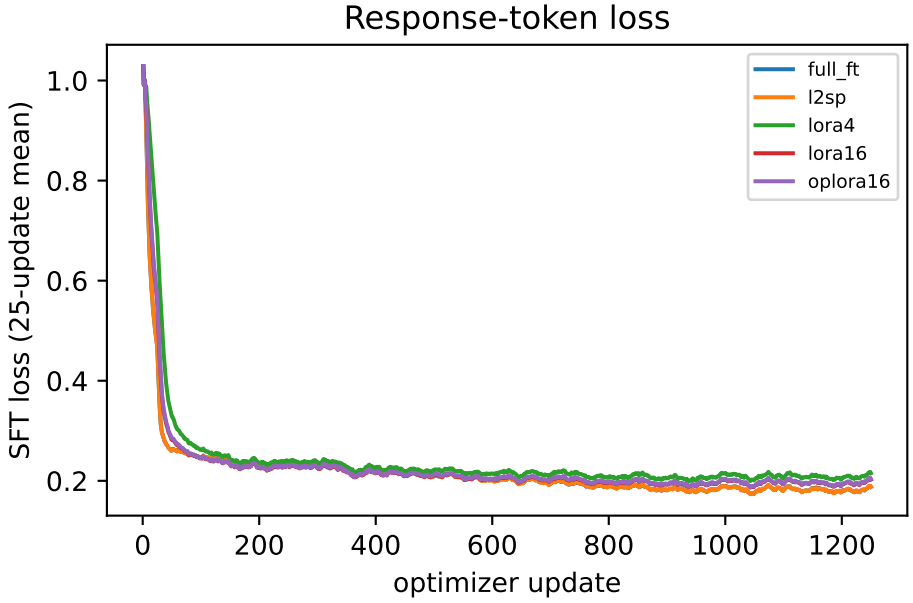
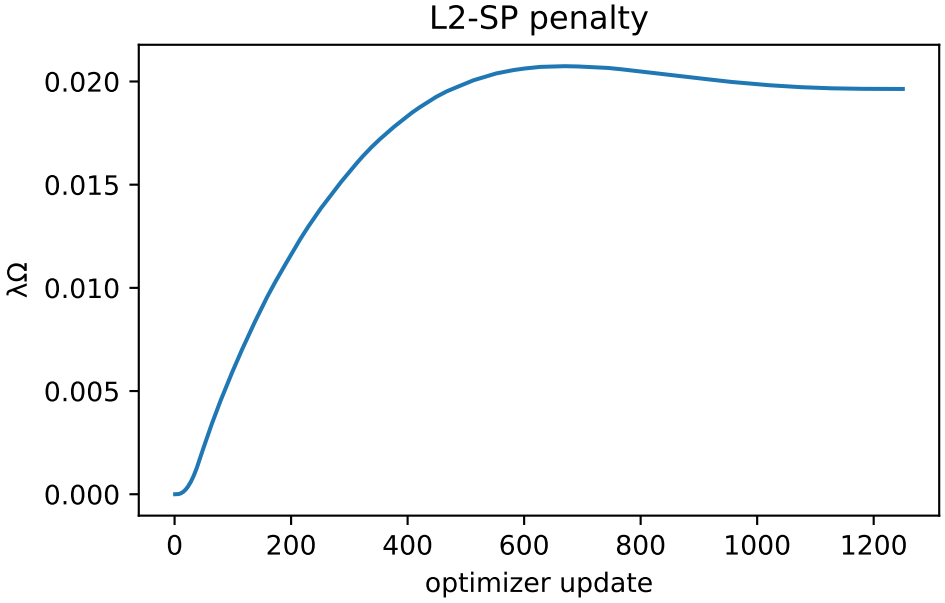

Response-token loss
SFT loss (25-update mean)
optimizer update
full_ft
l2sp
lora4
lora16
oplora16
1.0
0.8
0.6
0.4
0.2
0
200
400
600
800
1000
1200


L2-SP penalty
λΩ
optimizer update
0.020
0.015
0.010
0.005
0.000
0
200
400
600
800
1000
1200